# Public Economics (ECON 131)
# Section #4: Efficiency Costs of Taxation

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## 1 Taxation and Efficiency

The Second Welfare Theorem tells us that the ideal way to tax somebody (moral perceptions of fairness aside) is based on their innate characteristics–i.e. things that are impossible for them to change with their behavior–but, this is easier said than done. Suppose we tried to enact an income tax based on people's *potential* earnings. The government would somehow have to know how skilled the person is at every job, measuring things from intelligence to health to interpersonal skills, and to figure out how these things would translate into the labor market.

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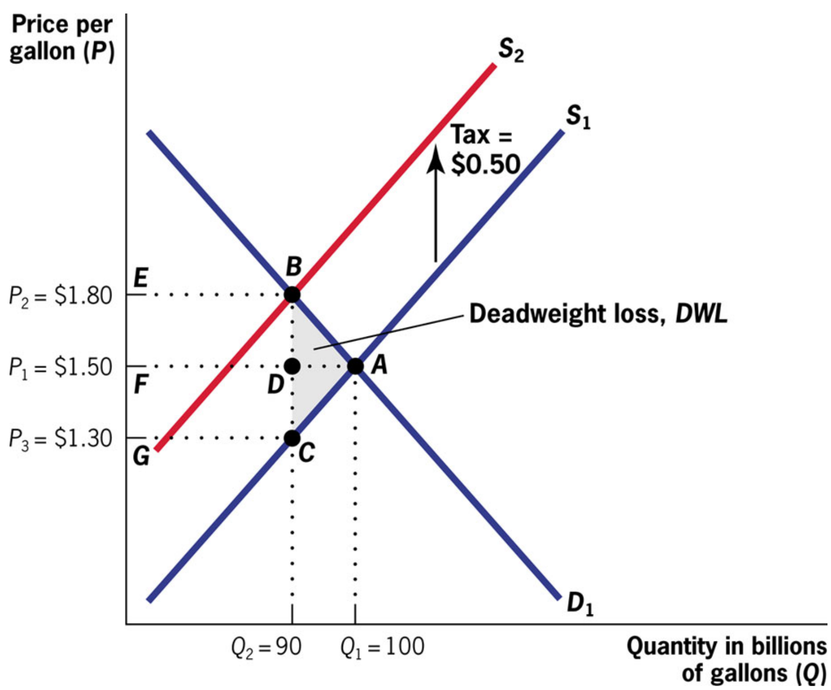
Instead of doing all that, the government imposes taxes based on *outcomes*, that is, how much money you actually make, what you purchase and how well your investments do. When we tax things that people can control, we risk distorting incentives in that market. If we tax wages, for instance, people might choose to work more or less than is optimal. People can go to great lengths to avoid taxes. One dramatic instance of tax avoidance with negative consequences is documented in Oates and Schwab (2015) and retold in the Gruber textbook. In 1696, King William III of England instituted a tax based on the number of windows in a house. In response, people boarded up and closed off windows, reducing ventilation, and aiding the spread of disease. Other, more modern examples of tax avoidance include ABBA dressing oddly to write-off their clothing purchases and White Claw being fermented like a beer (in part) to avoid the higher taxes associated with vodka-sodas.

Of course, taxation can also be used to fix distortions that already exist in the market. For instance, many economists and environmentalists advocate for a carbon tax. In this case, the price of carbon-intensive goods and services would increase to better reflect the externalizes these products have on the health of the planet and we who inhabit it. In other words, taxes could make the market function *more* efficiently.

**Deadweight burden** (DWB, also called excess burden or deadweight loss (DWL)) of taxation is defined as the welfare loss (measured in dollars) created by a tax, relative to the baseline welfare in the free market.

In a simple supply and demand diagram, welfare is measured by the sum of the consumer surplus and producer surplus. We can measure the DWL of taxation as the change in consumer + producer surplus minus tax collected: it is the triangle on the figure, which economists call the **Harberger Triangle**:




Mathematically, area of this triangle is given by:

$$DWB = \frac{1}{2}dQ \cdot dt = \frac{1}{2} \cdot \frac{\varepsilon_S \cdot \varepsilon_D}{\varepsilon_S - \varepsilon_D} \cdot \frac{Q}{p}(dt)^2$$

Looking at the formula, we can see that if there is no change in quantities consumed, the tax has no efficiency costs. The inefficiency of any tax is determined by the extent to which consumers and producers change their behavior to avoid the tax. Deadweight loss is caused by individuals and firms making inefficient consumption and production choices in order to avoid taxation.

We can also observe that the deadweight loss increases with the square of the tax, meaning that a tax twice as big will result in a deadweight loss that is four times as big.

These observations lead to two important rules of thumb for reducing the efficiency costs of a tax:

1. *The Inverse Elasticity Rule:* The more inelastic the demand for a good is, the more it should be taxed

2. *The Base-Broadening Rule:* It is better to tax a wide variety of goods at a moderate rate than to tax very few goods at a high rate.

Of course, as rules of thumb, there are many exceptions. Furthermore, these rules do not take into account any distributional or fairness concerns. For instance, suppose the government could only tax caviar or insulin. The demand for caviar is likely more elastic, so the implication of these rules is that you ought to tax insulin at a higher rate than caviar. However, many people would view that system as unjust. In practice, many essential commodities face lower tax rates. For instance, the California state sales tax does not apply to groceries or prescription medication.

## 2 Practice problems

### 2.1 Gruber, Ch.20, Q.2

The government of Washlovia wants to impose a tax on clothes dryers. In East Washlovia the demand elasticity for clothes dryers is -2.4 while in West Washlovia the demand elasticity is -1.7. Where will the tax inefficiency be greater for the same level tax? Explain.

**Solution:**

- The more elastic the demand, the more inefficient the tax (because the more people distort their behavior from their optimum); therefore, the tax will be less efficient in East Washlovia.
- Tax inefficiency is measured by the area of deadweight loss it generates.
- The base of the deadweight loss triangle will be the same in both regions of Washlovia, since the dollar amount of the tax is the same.
- The height of the deadweight loss triangle will differ, because it is the quantity change in response to the tax.
- Elasticity is a measure of quantity response to a price change: the higher the elasticity, the greater the quantity change for a given price change. When demand is elastic, a price change will distort quantity demanded by relatively more than when it is inelastic, and it is this quantity distortion that causes inefficiency.

### 2.2 Gruber, Ch.20, Q.10 (adapted)

The market demand for pet turtles is $Q = 2,600 - 20P$, and the government intends to place a \$4 per turtle tax on pet turtle purchases. Calculate the deadweight loss of this tax when:

1. Supply of pet turtles is $Q = 400$.
2. Supply of pet turtles is $Q = 12P$.
3. Explain why the deadweight loss calculations differ between 1 and 2.

**Solution:**

1. The quantity before the tax is 400; the quantity after the tax is 400. Quantity is completely inelastic. When supply is always 400 turtles, the deadweight loss of the turtle tax is $\frac{1}{2} \cdot (4 * 0)$, or 0. There is no change in supply, so there is no deadweight loss.
2. In this case, supply is not completely inelastic, so before-tax and after-tax quantities must be calculated.
   - Before tax:
   $$2,600 - 20P = 12P$$
   $$\Rightarrow P = \$81.25; \quad Q = 12 * 81.25 = 975$$
   $2,600 - 20P = 12P \quad \Rightarrow \quad P = \$81.25; Q = 12 * 81.25 = 975.$
   - After tax:
   $$2,600 - 20P = 12(P - 4) \Rightarrow$$
   $$2,600 - 20P = 12P - 48 \Rightarrow$$
   $$2,648 = 32P \Rightarrow$$
   $$P = 82.75; \quad Q = 2,600 - (20 * 82.75) = 945$$
   - The quantity change is $975 - 945 = 30$, so the area of the deadweight loss triangle is $\frac{1}{2}(30 \cdot 4) = 60$.
3. Deadweight loss is caused by changes in the equilibrium quantity. In (1), because supply was perfectly inelastic, there was no change in quantity. When quantity does not change, the tax has caused no distortion. Thus, there is no deadweight loss, only a transfer of money from the seller to the government.